

# EMS

## Employee Management System

Frontend Rebuild Blueprint

Complete API Reference · Data Models · Bolt/v0 Prompt · Architecture

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Next.js 15 · TypeScript · TailwindCSS · shadcn/ui · Socket.IO

# 1. Project Overview

Full-stack Employee Management System — Frontend Architecture

This document is a **complete rebuild specification** for the EMS frontend. It contains every API endpoint, data model field, component list, environment variable, route configuration, and a ready-to-paste prompt for Bolt.new or v0.dev to regenerate the entire application from scratch.

| Property         | Value                                                     |
|------------------|-----------------------------------------------------------|
| Project Name     | employee-management-system                                |
| Framework        | Next.js 15.0.7 (App Router)                               |
| Language         | TypeScript 5.x                                            |
| Styling          | TailwindCSS 4.x + shadcn/ui + Radix UI                    |
| State Management | React Context API (Auth, Toast, Attendance, TokenExpiry)  |
| Real-time        | Socket.IO Client 4.8.1                                    |
| Forms            | React Hook Form 7.x + Zod 3.x validation                  |
| Charts           | Recharts 2.15                                             |
| Animations       | Framer Motion 12.x + GSAP 3.x                             |
| DnD              | @hello-pangea/dnd 16.x (Kanban board)                     |
| Package Manager  | pnpm (also npm-compatible)                                |
| API Base URL     | http://localhost:3005/api (env: NEXT_PUBLIC_API_BASE_URL) |
| Socket URL       | http://localhost:3005 (env: NEXT_PUBLIC_API_URL)          |
| Auth Method      | Bearer JWT token — stored in localStorage as 'authToken'  |
| User Store       | localStorage key: 'user' (JSON-serialized user object)    |

# 2. Pages, Routes & Role Access

The application uses Next.js App Router. All pages below live in app/ directory. Role-based visibility is enforced in the sidebar component.

| Route            | File                         | Access Roles       | Description                           |
|------------------|------------------------------|--------------------|---------------------------------------|
| /                | app/page.tsx                 | Public             | Login page with credential form       |
| /forgot-password | app/forgot-password/page.tsx | Public             | Send OTP to email for reset           |
| /verify-otp      | app/verify-otp/page.tsx      | Public             | OTP verification (6-digit)            |
| /reset-password  | app/reset-password/page.tsx  | Public             | Set new password after OTP verify     |
| /dashboard       | app/dashboard/page.tsx       | All roles          | Overview stats, charts, quick actions |
| /profile         | app/profile/page.tsx         | All roles          | Edit own profile, avatar upload       |
| /employees       | app/employees/page.tsx       | admin, hr, manager | Full employee CRUD, search, filter    |
| /add-user        | app/add-user/page.tsx        | admin, hr          | Register new employee form            |

|              |                          |                           |                                       |
|--------------|--------------------------|---------------------------|---------------------------------------|
| /projects    | app/projects/page.tsx    | admin, manager, developer | Project CRUD, team assignment, Kanban |
| /tasks       | app/tasks/page.tsx       | admin, manager, developer | Kanban board (drag-and-drop tasks)    |
| /performance | app/performance/page.tsx | admin only                | Employee performance analytics        |
| /attendance  | app/attendance/page.tsx  | All roles                 | Punch-in/out, attendance logs & stats |
| /leave       | app/leave/page.tsx       | All roles                 | Apply, approve, reject leave requests |
| /work-logs   | app/work-logs/page.tsx   | All roles                 | Daily work log entries                |
| /support     | app/support/page.tsx     | All roles                 | Support ticket + real-time chat       |
| /team-inbox  | app/team-inbox/page.tsx  | All roles                 | Shared team message board             |
| /holidays    | app/holidays/page.tsx    | All roles                 | Holiday calendar with sidebar         |
| /policies    | app/policies/page.tsx    | All roles                 | PDF policy documents library          |
| /emails      | app/emails/page.tsx      | admin, manager            | Email records management              |
| /reports     | app/reports/page.tsx     | admin, hr                 | Performance export reports            |
| /settings    | app/settings/page.tsx    | admin, hr                 | App & notification settings           |
| /activity    | app/activity/page.tsx    | admin, hr, manager        | User activity logs                    |

# 3. Complete API Reference

Base URL: `http://localhost:3005/api` | Auth: Bearer header on all protected routes

## 3.1 Authentication API

|        |                                                   |                                                                               |
|--------|---------------------------------------------------|-------------------------------------------------------------------------------|
| POST   | <code>/user/auth/login</code>                     | Login — body: { identifier, password }. Returns { status, token, data: User } |
| POST   | <code>/user/auth/register</code>                  | Register new user — body: UserPayload (see model). Admin/HR only.             |
| GET    | <code>/user/auth/users</code>                     | List all users. Params: q, role, department, page, limit                      |
| GET    | <code>/user/auth/users/:id</code>                 | Get single user by ID                                                         |
| PUT    | <code>/user/auth/users/:id</code>                 | Update user fields by ID                                                      |
| DELETE | <code>/user/auth/users/:id</code>                 | Delete user by ID. Admin only.                                                |
| POST   | <code>/user/auth/users/:id/toggle-disabled</code> | Toggle user active/disabled status                                            |
| PUT    | <code>/user/auth/update-profile</code>            | Update own profile. Body: ProfilePayload                                      |
| POST   | <code>/user/auth/upload-avatar</code>             | Upload avatar image. FormData: field 'image'. Returns { avatar_url }          |
| GET    | <code>/user/auth/getActivityLogs</code>           | Get all activity logs. Params: page, limit                                    |
| GET    | <code>/user/auth/getActivityLogsById</code>       | Get activity logs for user. Params: id (required), page, limit                |

## 3.2 Password Reset API

|      |                                             |                                                                    |
|------|---------------------------------------------|--------------------------------------------------------------------|
| POST | <code>/password-reset/send-otp</code>       | Send OTP to email. Body: { email }                                 |
| POST | <code>/password-reset/resend-otp</code>     | Resend OTP. Body: { email }                                        |
| POST | <code>/password-reset/verify-otp</code>     | Verify OTP. Body: { email, otp }                                   |
| POST | <code>/password-reset/reset-password</code> | Reset password. Body: { email, otp, newPassword, confirmPassword } |

## 3.3 Projects API

|        |                                 |                                                            |
|--------|---------------------------------|------------------------------------------------------------|
| GET    | <code>/user/projects</code>     | List projects. Params: page, pageSize, q, status, priority |
| POST   | <code>/user/projects</code>     | Create project. Body: ProjectPayload                       |
| GET    | <code>/user/projects/:id</code> | Get single project                                         |
| PUT    | <code>/user/projects/:id</code> | Update project. Body: Partial                              |
| DELETE | <code>/user/projects/:id</code> | Delete project                                             |

|        |                                     |                                                                |
|--------|-------------------------------------|----------------------------------------------------------------|
| POST   | /user/projects/:id/attachments      | Add attachments. Body: { files: string[] } or { file: string } |
| DELETE | /user/projects/:id/attachments/:idx | Remove attachment by index or filename                         |
| POST   | /user/projects/upload-image         | Upload image file. FormData field: 'image'                     |
| POST   | /user/projects/upload-file          | Upload document file. FormData field: 'file'                   |
| POST   | /user/projects/upload-audio         | Upload audio file. FormData field: 'audio'                     |
| POST   | /user/projects/upload-video         | Upload video file. FormData field: 'video'                     |
| GET    | /user/projects/file/:filename       | Serve project file (token via query ?token=)                   |

### 3.4 Tasks API

|        |                                                  |                                                                                                                          |
|--------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| GET    | /user/tasks                                      | List tasks. Params: status, priority, projectId, project, page, pageSize, q, user, startDate, endDate, sortBy, sortOrder |
| POST   | /user/tasks                                      | Create task. Body: TaskPayload                                                                                           |
| GET    | /user/tasks/:id                                  | Get single task                                                                                                          |
| PUT    | /user/tasks/:id                                  | Update task. Body: Partial                                                                                               |
| DELETE | /user/tasks/:id                                  | Delete task                                                                                                              |
| PUT    | /user/tasks/:id/complete                         | Mark complete. Body: { done: boolean }                                                                                   |
| PUT    | /user/tasks/:id/complete-user                    | Per-user completion. Body: { userId, completed }                                                                         |
| PUT    | /user/tasks/:id/progress                         | Update progress. Body: { progress: 0-100 }                                                                               |
| PUT    | /user/tasks/:id/approve                          | Approve task. Body: { approval_notes? }                                                                                  |
| PUT    | /user/tasks/:id/unapprove                        | Unapprove task                                                                                                           |
| PUT    | /user/tasks/:id/hold                             | Toggle hold. Body: { hold_task, reason? }                                                                                |
| POST   | /user/tasks/:id/subtasks                         | Add subtask. Body: SubtaskPayload                                                                                        |
| PUT    | /user/tasks/:id/subtasks/:subId                  | Update subtask                                                                                                           |
| DELETE | /user/tasks/:id/subtasks/:subId                  | Delete subtask                                                                                                           |
| POST   | /user/tasks/:id/subtasks/:subId/attachments      | Add subtask attachments                                                                                                  |
| DELETE | /user/tasks/:id/subtasks/:subId/attachments/:idx | Remove subtask attachment                                                                                                |

|        |                                     |                                              |
|--------|-------------------------------------|----------------------------------------------|
| POST   | /user/tasks/:id/attachments         | Add task attachments                         |
| DELETE | /user/tasks/:id/attachments/:idx    | Remove task attachment                       |
| POST   | /user/tasks/:id/comments            | Add comment. Body: { content, attachments? } |
| DELETE | /user/tasks/:id/comments/:commentId | Delete comment                               |
| GET    | /user/tasks/:id/comments            | List task comments                           |
| POST   | /user/tasks/upload-image            | Upload image. FormData: 'image'              |
| POST   | /user/tasks/upload-file             | Upload file. FormData: 'file'                |
| POST   | /user/tasks/upload-audio            | Upload audio. FormData: 'audio'              |
| POST   | /user/tasks/upload-video            | Upload video. FormData: 'video'              |
| GET    | /user/tasks/file/:filename          | Serve task file (token via ?token=)          |

## 3.5 Attendance API

|      |                                  |                                               |
|------|----------------------------------|-----------------------------------------------|
| GET  | /user/attendance/status          | Get user punch status. Param: user_id         |
| POST | /user/attendance/punch-in        | Punch in. Body: { user_id }                   |
| POST | /user/attendance/punch-out       | Punch out. Body: { user_id }                  |
| GET  | /user/attendance/stats/today     | Today's attendance stats (admin/HR)           |
| GET  | /user/attendance/today           | Today's attendance list                       |
| POST | /user/attendance/check-in        | Check in (own user, no body needed)           |
| POST | /user/attendance/check-out       | Check out (own user, no body needed)          |
| GET  | /user/attendance/by-date         | Attendance by date. Param: date (ISO)         |
| GET  | /user/attendance/range           | Date range attendance. Params: from, to (ISO) |
| GET  | /user/attendance/user/:userId    | User history. Params: from, to                |
| GET  | /user/attendance/my-stats        | Own attendance stats                          |
| GET  | /user/attendance/my-weekly-hours | Own weekly hours breakdown                    |

## 3.6 Leave API

|      |                     |                                                                                            |
|------|---------------------|--------------------------------------------------------------------------------------------|
| GET  | /leaves/my          | Own leave requests. Params: page, limit, q, month                                          |
| GET  | /leaves/all         | All leaves (admin/HR). Params: page, limit, q, userId, overviewUserId, status, date, month |
| POST | /leaves/apply       | Apply for leave. Body: { dateFrom, dateTo, reason, type?, duration? }                      |
| PUT  | /leaves/:id/approve | Approve leave request                                                                      |
| PUT  | /leaves/:id/reject  | Reject leave. Body: { cancellation_reason }                                                |
| PUT  | /leaves/:id/cancel  | Cancel leave. Body: { cancellation_reason }                                                |

## 3.7 Support Tickets & Chat API

|      |                     |                                                                                    |
|------|---------------------|------------------------------------------------------------------------------------|
| GET  | /support            | List tickets. Params: page, pageSize, status, employeeId, category, assigned_to, q |
| POST | /support            | Create ticket. Body: SupportTicketPayload                                          |
| GET  | /support/:id        | Get single ticket                                                                  |
| PUT  | /support/:id        | Update ticket. Body: Partial                                                       |
| PUT  | /support/:id/assign | Assign ticket. Body: { assigned_to }                                               |

|        |                             |                                                                    |
|--------|-----------------------------|--------------------------------------------------------------------|
| DELETE | /support/:id                | Delete ticket                                                      |
| GET    | /chat/:ticketId             | Get messages for ticket                                            |
| POST   | /chat                       | Send text message. Body: { ticket_id, message, username, type }    |
| POST   | /chat/upload-image          | Upload image. FormData: image, ticket_id, message?, type, username |
| POST   | /chat/upload-file           | Upload file. FormData: file, ticket_id, message?, type, username   |
| POST   | /chat/upload-audio          | Upload audio. FormData: audio, ticket_id, message?, type, username |
| POST   | /chat/upload-video          | Upload video. FormData: video, ticket_id, message?, type, username |
| GET    | /chat/file/:filename        | Serve chat attachment. Param: ticket_id                            |
| DELETE | /chat/:messageId/attachment | Remove attachment from message                                     |
| DELETE | /chat/:messageId            | Delete message for everyone                                        |
| GET    | /chat                       | Get all messages (admin). Params: ticket_id, username, page, limit |

### 3.8 Team Inbox / Work Logs / Reports / Settings / Misc

|        |                               |                                                                     |
|--------|-------------------------------|---------------------------------------------------------------------|
| GET    | /team-inbox                   | List team messages. Params: page, limit                             |
| POST   | /team-inbox                   | Send message. Body: { message }                                     |
| GET    | /team-inbox/:id               | Get single message                                                  |
| PUT    | /team-inbox/:id               | Update message. Body: { message }                                   |
| DELETE | /team-inbox/:id               | Delete message                                                      |
| GET    | /work-logs                    | List work logs. Params: user_id, from, to, period                   |
| POST   | /work-logs                    | Create work log. Body: CreateWorkLogPayload                         |
| GET    | /reports/employee-performance | Employee perf report. Params: period, department, role, performance |
| GET    | /reports/performance-summary  | Summary stats. Params: period                                       |
| GET    | /reports/export-performance   | Export report. Params: format, period                               |
| GET    | /settings                     | Get all settings                                                    |
| PUT    | /settings                     | Update all settings. Body: SettingsData                             |
| GET    | /settings/:section            | Get settings section                                                |
| PUT    | /settings/:section            | Update settings section. Body: section data                         |
| GET    | /holidays                     | List holidays. Params: from, to (ISO dates)                         |
| POST   | /holidays                     | Create holiday. Body/FormData: date, title, description, image      |



|        |               |                                                            |
|--------|---------------|------------------------------------------------------------|
| PUT    | /holidays/:id | Update holiday. Body/FormData: date?, title?, description? |
| DELETE | /holidays/:id | Delete holiday                                             |
| GET    | /emails       | List email records. Params: page, limit, q                 |
| GET    | /emails/:id   | Get single email record                                    |
| PUT    | /emails/:id   | Update email. Body: { name?, email? }                      |
| GET    | /policies     | List policy documents (admin/HR fetch)                     |
| POST   | /policies     | Create policy. FormData: name, file, assigned_user_id?     |
| DELETE | /policies/:id | Delete policy document                                     |

### 3.9 Task & Ticket OTP API (Next.js API routes + proxy)

|      |                        |                                                       |
|------|------------------------|-------------------------------------------------------|
| POST | /task-otp/send-otp     | Send OTP for task approval. Body: { task_id, email? } |
| POST | /task-otp/verify-otp   | Verify task OTP. Body: { task_id, email, otp }        |
| POST | /task-otp/resend-otp   | Resend task OTP. Body: { task_id, email? }            |
| GET  | /task-otp/recipient    | Get OTP recipient. Param: task_id                     |
| POST | /ticket-otp/send-otp   | Send OTP for ticket. Body: { ticket_id }              |
| POST | /ticket-otp/verify-otp | Verify ticket OTP. Body: { ticket_id, otp }           |
| POST | /ticket-otp/resend-otp | Resend ticket OTP. Body: { ticket_id }                |

## 4. Complete Data Models

All TypeScript interfaces — exact field names, types, and constraints

### 4.1 User / Auth Models

| Field         | Type    | Req | Notes                                            |
|---------------|---------|-----|--------------------------------------------------|
| id            | number  | ✓   | Auto-incremented primary key                     |
| fullname      | string  | ✓   | Display name                                     |
| email         | string  | ✓   | Unique email address                             |
| phone         | string  | –   | Phone number                                     |
| password      | string  | ✓   | Hashed on server; sent on register/update        |
| role          | string  | ✓   | Enum: admin   hr   manager   developer           |
| department    | string  | –   | Department name                                  |
| address       | string  | –   | Full address                                     |
| bio           | string  | –   | Short biography                                  |
| avatar_url    | string  | –   | Profile image URL (after upload)                 |
| website_url   | string  | –   | Personal/company website                         |
| linkedin_url  | string  | –   | LinkedIn profile URL                             |
| github_url    | string  | –   | GitHub profile URL                               |
| date_of_birth | string  | –   | ISO date string YYYY-MM-DD                       |
| employee_id   | string  | –   | Human-readable employee code (e.g. EMP001)       |
| is_disabled   | boolean | –   | Account disabled flag (toggle-disabled endpoint) |

### 4.2 Project Model

| Field       | Type     | Req | Notes                                           |
|-------------|----------|-----|-------------------------------------------------|
| name        | string   | ✓   | Project title                                   |
| description | string   | –   | Project description                             |
| status      | string   | –   | Enum: planning   active   completed   cancelled |
| priority    | string   | –   | Enum: high   medium   low                       |
| progress    | number   | –   | 0–100 percent completion                        |
| team        | string[] | –   | Array of user IDs or names assigned             |
| startDate   | string   | –   | ISO date YYYY-MM-DD                             |
| dueDate     | string   | –   | ISO date YYYY-MM-DD                             |
| budget      | number   | –   | Project budget amount                           |
| spent       | number   | –   | Amount spent so far                             |
| client      | string   | –   | Client name                                     |
| category    | string   | –   | Project category/type                           |
| tasks.total | number   | –   | Total task count                                |

|                      |          |   |                           |
|----------------------|----------|---|---------------------------|
| tasks.comple<br>ted  | number   | – | Completed task count      |
| tasks.inProg<br>ress | number   | – | In-progress task count    |
| tasks.pendin<br>g    | number   | – | Pending task count        |
| attachments          | string[] | – | Array of server filenames |

### 4.3 Task Model

| Field                | Type                 | Re<br>q | Notes                                    |
|----------------------|----------------------|---------|------------------------------------------|
| title                | string               | ✓       | Task title                               |
| description          | string               | –       | Detailed description                     |
| status               | string               | –       | Enum: todo   in-progress   review   done |
| priority             | string               | –       | Enum: high   medium   low                |
| assignee             | string               | –       | Assigned user name or ID                 |
| project              | string               | –       | Project name (display)                   |
| projectId            | string n<br>umber    | –       | Project foreign key                      |
| startDate            | string               | –       | ISO date YYYY-MM-DD                      |
| startTime            | string               | –       | HH:mm format                             |
| deadlineDate         | string               | –       | ISO date YYYY-MM-DD                      |
| dueDate              | string               | –       | Alternative deadline field               |
| progress             | number               | –       | 0–100 percent                            |
| review_count         | number               | –       | Times moved to review                    |
| completed_by         | string               | –       | User who completed                       |
| completed_at         | string               | –       | ISO datetime of completion               |
| overtime_day<br>s    | number               | –       | Days over deadline                       |
| hold_task            | boolean              | –       | Task is on hold                          |
| approved             | boolean              | –       | Approved by manager                      |
| subtasks             | Subtask[<br>]        | –       | Array of subtask objects                 |
| comments             | Comment[<br>]        | –       | Array of comment objects                 |
| attachments          | string[]             | –       | Array of server filenames                |
| status_histo<br>ry   | StatusHi<br>story[]  | –       | Audit trail of status changes            |
| user_complet<br>ions | UserComp<br>letion[] | –       | Per-user completion tracking             |

Subtask fields:

| Field | Type   | Re<br>q | Notes                              |
|-------|--------|---------|------------------------------------|
| id    | string | –       | UUID generated on client or server |
| title | string | ✓       | Subtask title                      |

|             |          |   |                  |
|-------------|----------|---|------------------|
| description | string   | – | Description      |
| completed   | boolean  | – | Completion flag  |
| assignee    | string   | – | Assigned user    |
| attachments | string[] | – | Server filenames |
| createdAt   | string   | – | ISO datetime     |
| updatedAt   | string   | – | ISO datetime     |

Comment fields:

| Field       | Type     | Req | Notes            |
|-------------|----------|-----|------------------|
| id          | string   | –   | UUID             |
| author      | string   | ✓   | Author name      |
| content     | string   | ✓   | Message text     |
| timestamp   | string   | ✓   | ISO datetime     |
| attachments | string[] | –   | Server filenames |

## 4.4 Attendance Models

| Field       | Type            | Re<br>q | Notes                                    |
|-------------|-----------------|---------|------------------------------------------|
| id          | number   string | ✓       | Record ID                                |
| employee    | string          | ✓       | Employee full name                       |
| date        | string          | ✓       | ISO date YYYY-MM-DD                      |
| checkIn     | string          | ✓       | Time string HH:mm or ISO                 |
| checkOut    | string          | ✓       | Time string HH:mm or ISO                 |
| status      | string          | ✓       | Enum: present   absent   late   half-day |
| hoursWorked | number          | ✓       | Decimal hours e.g. 8.5                   |
| userId      | number          | ✓       | Foreign key to User                      |

AttendanceStats object:

| Field          | Type   | Re<br>q | Notes               |
|----------------|--------|---------|---------------------|
| totalEmployees | number | ✓       | Total headcount     |
| presentToday   | number | ✓       | Present today count |
| absentToday    | number | ✓       | Absent today count  |
| lateToday      | number | ✓       | Late today count    |
| averageHours   | number | ✓       | Avg hours worked    |

## 4.5 Leave Model

| Field               | Type   | Re<br>q | Notes                                     |
|---------------------|--------|---------|-------------------------------------------|
| dateFrom            | string | ✓       | ISO date YYYY-MM-DD                       |
| dateTo              | string | ✓       | ISO date YYYY-MM-DD                       |
| reason              | string | ✓       | Leave reason text                         |
| type                | string | –       | Leave type: casual   sick   annual   etc. |
| duration            | string | –       | full   half-day                           |
| status              | string | –       | pending   approved   rejected   cancelled |
| cancellation_reason | string | –       | Reason for reject/cancel                  |
| userId              | number | –       | FK — set by server from JWT               |

## 4.6 Support Ticket Model

| Field        | Type   | Re<br>q | Notes                        |
|--------------|--------|---------|------------------------------|
| username     | string | ✓       | Requester username           |
| category     | string | –       | Ticket category              |
| ticket_id    | string | –       | Human-readable ticket number |
| sub_category | string | –       | Sub-category                 |

|                      |          |   |                                                    |
|----------------------|----------|---|----------------------------------------------------|
| subject_sub_category | string   | – | Subject sub-category                               |
| subject              | string   | ✓ | Ticket subject                                     |
| body                 | string   | ✓ | Full description                                   |
| media                | string[] | – | Array of attachment filenames                      |
| assigned_to          | string   | – | Assignee user name/ID                              |
| status               | 0 1 2 3  | – | 0=Pending, 1=Open, 2=Closed, 3=Under Investigation |
| reseller             | string   | – | Reseller info if applicable                        |
| message_by           | string   | – | Message sender identifier                          |

## 4.7 Chat Message Model

| Field           | Type   | Req | Notes                               |
|-----------------|--------|-----|-------------------------------------|
| id              | number | ✓   | Auto-incremented ID                 |
| username        | string | ✓   | Sender's display name               |
| message         | string | ✓   | Message text                        |
| ticket_id       | string | ✓   | Parent ticket ID                    |
| type            | string | ✓   | Enum: CustomerSupport   User        |
| created_at      | string | ✓   | ISO datetime                        |
| attachment_type | string | –   | none   image   file   audio   video |
| attachment_name | string | –   | Original filename                   |
| attachment_path | string | –   | Server-stored filename              |
| attachment_size | number | –   | File size in bytes                  |

## 4.8 Work Log Model

| Field        | Type   | Req | Notes                     |
|--------------|--------|-----|---------------------------|
| id           | number | ✓   | Auto ID                   |
| user_id      | number | ✓   | FK to User                |
| date         | string | ✓   | ISO date YYYY-MM-DD       |
| project_name | string | –   | Project name              |
| task         | string | ✓   | Task description          |
| blockers     | string | –   | Any blockers              |
| description  | string | –   | Additional notes          |
| time_spent   | number | –   | Hours (decimal) default 0 |
| status       | string | –   | Log status                |

## 4.9 Holiday Model

| Field       | Type   | Req | Notes                             |
|-------------|--------|-----|-----------------------------------|
| id          | number | ✓   | Auto ID                           |
| date        | string | ✓   | ISO date YYYY-MM-DD               |
| title       | string | ✓   | Holiday name                      |
| description | string | –   | Optional description              |
| image_path  | string | –   | Server filename for holiday image |
| created_by  | number | –   | FK to User who created it         |
| created_at  | string | –   | ISO datetime                      |
| updated_at  | string | –   | ISO datetime                      |

## 4.10 Policy Model (frontend-only)

| Field        | Type   | Req | Notes                                        |
|--------------|--------|-----|----------------------------------------------|
| id           | string | ✓   | Server-assigned ID                           |
| name         | string | ✓   | Policy document name                         |
| fileName     | string | ✓   | Stored filename on server                    |
| fileSize     | number | ✓   | Size in bytes                                |
| uploadedAt   | string | ✓   | ISO datetime                                 |
| uploadedBy   | string | ✓   | Uploader name                                |
| fileUrl      | string | ✓   | Object URL (frontend preview only)           |
| assignedUser | object | –   | { id, fullname, email, employee_id? }   null |

# 5. Component Architecture

## 5.1 Provider / Context Hierarchy

The root layout wraps the app in this exact provider order:

ErrorBoundary > TokenExpiryProvider > AuthProvider > ToastProvider > AttendanceProvider > ClientLayout

## 5.2 Context Files in lib/

| File                                     | Exports / Provides                    | Description                                                          |
|------------------------------------------|---------------------------------------|----------------------------------------------------------------------|
| <a href="#">auth-context.tsx</a>         | AuthProvider, useAuth()               | JWT auth: user, token, login(), logout(), isAuthenticated, isLoading |
| <a href="#">toast-context.tsx</a>        | ToastProvider, useToast()             | App-wide toast notifications                                         |
| <a href="#">attendance-context.tsx</a>   | AttendanceProvider, useAttendance()   | Punch-in/out state, today's status                                   |
| <a href="#">token-expiry-context.tsx</a> | TokenExpiryProvider, useTokenExpiry() | Shows modal on 401 response                                          |

## 5.3 Shared Components (components/)

| Component               | File                                          | Description                                            |
|-------------------------|-----------------------------------------------|--------------------------------------------------------|
| Sidebar                 | <a href="#">sidebar.tsx</a>                   | Left nav with role-based links, collapse, active state |
| MobileSidebar           | <a href="#">mobile-sidebar.tsx</a>            | Sheet-based mobile nav drawer                          |
| HeaderAttendance        | <a href="#">header-attendance.tsx</a>         | Top header with punch-in/out button                    |
| AttendanceTracker       | <a href="#">attendance-tracker.tsx</a>        | Full attendance view with calendar                     |
| TaskManagement          | <a href="#">task-management.tsx</a>           | Kanban board with DnD, subtasks, comments              |
| SubtaskManager          | <a href="#">subtask-manager.tsx</a>           | Subtask list with CRUD operations                      |
| SubtaskDialog           | <a href="#">subtask-dialog.tsx</a>            | Modal for creating/editing subtasks                    |
| EnhancedSubtask         | <a href="#">enhanced-subtask.tsx</a>          | Rich subtask with attachments                          |
| ProjectOverview         | <a href="#">project-overview.tsx</a>          | Project detail panel with stats                        |
| ProjectAttachmentUpload | <a href="#">project-attachment-upload.tsx</a> | Multi-type file uploader for projects                  |
| OnlineMembers           | <a href="#">online-members.tsx</a>            | Shows online users (Socket.IO presence)                |
| TeamInbox               | <a href="#">team-inbox.tsx</a>                | Shared team message board UI                           |
| SupportChat             | <a href="#">support-chat.tsx</a>              | Real-time chat window for support tickets              |
| LeaveComponent          | <a href="#">leave/page.tsx</a>                | Full leave management (apply/approve/reject)           |
| HolidaysManagement      | <a href="#">holidays-management.tsx</a>       | Holiday CRUD with image upload                         |
| HolidaysSidebarCalendar | <a href="#">holidays-sidebar-calendar.tsx</a> | Calendar with holiday highlights                       |
| PoliciesManagement      | <a href="#">policies-management.tsx</a>       | PDF policy library with assignment                     |
| EmojiPicker             | <a href="#">emoji-picker.tsx</a>              | Custom emoji selector for chat                         |
| TicketOtpModal          | <a href="#">ticket-otp-modal.tsx</a>          | OTP verification modal for tickets                     |
| TaskOtpModal            | <a href="#">task-otp-modal.tsx</a>            | OTP verification modal for tasks                       |
| TokenExpiryModal        | <a href="#">token-expiry-modal.tsx</a>        | Session expired warning modal                          |
| ProtectedRoute          | <a href="#">protected-route.tsx</a>           | Wraps pages to enforce authentication                  |



|                     |                          |                                   |
|---------------------|--------------------------|-----------------------------------|
| BirthdayCelebration | birthday-celebration.tsx | Birthday confetti animation       |
| AnimatedBackground  | animated-background.tsx  | GSAP particle background          |
| ErrorBoundary       | error-boundary.tsx       | React error boundary wrapper      |
| TimeTracker         | time-tracker.tsx         | Live work-time tracker with timer |

## 6. Environment Variables & Configuration

| Variable                 | Default Value             | Purpose                                       |
|--------------------------|---------------------------|-----------------------------------------------|
| NEXT_PUBLIC_API_BASE_URL | http://localhost:3005/api | Backend REST API base URL                     |
| NEXT_PUBLIC_API_URL      | http://localhost:3005     | Backend origin (for Socket.IO + file serving) |

### Key Configuration Notes:

- Auth token is stored in `localStorage` under key `authToken`.
- Logged-in user object is stored in `localStorage` under key `user`.
- All API calls use `Authorization: Bearer <token>` header (injected in `apiRequest()`).
- 401 responses trigger the `TokenExpiryProvider` modal and do NOT redirect mid-request.
- Public routes (no auth required): `/` `/forgot-password` `/reset-password` `/verify-otp`.
- File uploads use XHR (not fetch) when progress tracking is needed — see `uploadFileWithProgress()`.
- Socket.IO connects to `NEXT_PUBLIC_API_URL` origin for real-time chat and presence.
- Tailwind v4 is used; config extends breakpoints from `xs` (475px) up to `desktop-lg` (1920px).
- Custom animations: `fade-in`, `slide-in`, `bounce-gentle`, `float`, `float-delay` (all defined in `tailwind.config.ts`).

# 7. Full Dependency List

## 7.1 Production Dependencies

| Package                  | Version  | Purpose                               |
|--------------------------|----------|---------------------------------------|
| next                     | 15.0.7   | React framework (App Router)          |
| react / react-dom        | ^18.0.0  | Core UI library                       |
| typescript               | ^5.0.0   | Type safety                           |
| tailwindcss              | ^4.1.9   | Utility-first CSS                     |
| @radix-ui/react-*        | various  | Headless UI primitives (20+ packages) |
| shadcn/ui (via Radix)    | —        | Pre-styled component system           |
| lucide-react             | ^0.454.0 | Icon set                              |
| @tabler/icons-react      | ^3.0.0   | Additional icons                      |
| framer-motion            | ^12.34.3 | Animations & transitions              |
| gsap                     | ^3.12.2  | Advanced animations (background)      |
| @hello-pangea/dnd        | ^16.6.0  | Drag-and-drop (Kanban)                |
| recharts                 | 2.15.4   | Charts & data visualization           |
| react-hook-form          | ^7.60.0  | Form state management                 |
| @hookform/resolvers      | ^3.10.0  | Zod integration for RHF               |
| zod                      | 3.25.67  | Schema validation                     |
| socket.io-client         | ^4.8.1   | WebSocket real-time client            |
| date-fns                 | 4.1.0    | Date utilities                        |
| react-day-picker         | 9.8.0    | Calendar picker component             |
| react-dropzone           | ^14.3.8  | Drag-and-drop file upload             |
| emoji-picker-react       | ^4.9.2   | Emoji selector                        |
| react-toastify           | 9.1.3    | Toast notification library            |
| sonner                   | ^1.7.4   | Alternative toast library             |
| next-themes              | ^0.4.6   | Dark/light theme provider             |
| geist                    | ^1.3.1   | Geist font for Next.js                |
| class-variance-authority | ^0.7.1   | CVA for variant styles                |
| clsx                     | ^2.1.1   | Conditional className utility         |
| tailwind-merge           | ^3.3.1   | Merge Tailwind class conflicts        |
| input-otp                | 1.4.1    | OTP digit input component             |
| react-resizable-panels   | ^2.1.7   | Resizable layout panels               |
| cmdk                     | 1.0.4    | Command palette component             |
| vaul                     | ^0.9.9   | Drawer component                      |
| embla-carousel-react     | 8.5.1    | Touch-friendly carousel               |
| @heroui/react            | ^2.8.4   | HeroUI component library              |

## 8. Bolt.new / v0.dev Master Rebuild Prompt

Paste this entire prompt directly into Bolt.new or v0.dev

The following is a production-ready prompt designed for AI-powered scaffolding tools. It contains all necessary context to regenerate the EMS frontend from scratch. Copy everything between the dashed lines.

Build a complete Employee Management System (EMS) frontend with Next.js 15 App Router, TypeScript, TailwindCSS 4, and shadcn/ui. Here is the full specification:

TECH STACK

Framework: Next.js 15.0.7 (App Router, no Pages Router)  
Language: TypeScript 5.x  
Styling: TailwindCSS 4.x + shadcn/ui components + Radix UI primitives  
Icons: lucide-react + @tabler/icons-react  
Animations: framer-motion + gsap  
Drag-and-Drop: @hello-pangea/dnd  
Charts: recharts  
Forms: react-hook-form + zod validation  
Real-time: socket.io-client  
Dates: date-fns  
Misc: react-dropzone, emoji-picker-react, react-toastify, sonner, next-themes

ENVIRONMENT VARIABLES (.env)

NEXT\_PUBLIC\_API\_BASE\_URL=http://localhost:3005/api  
NEXT\_PUBLIC\_API\_URL=http://localhost:3005

AUTHENTICATION

- JWT Bearer token, stored in localStorage as 'authToken'  
- User object stored in localStorage as 'user' (JSON)  
- All API calls inject: Authorization: Bearer  
- On 401 response: show token-expiry modal (do NOT redirect immediately)  
- Roles: admin | hr | manager | developer  
- Public routes: / (login), /forgot-password, /verify-otp, /reset-password

PAGES TO CREATE (app/ directory)

/ → Login page (identifier + password form)  
/forgot-password → Email OTP form  
/verify-otp → 6-digit OTP input  
/reset-password → New password form  
/dashboard → Stats cards + charts + quick actions  
/profile → Edit own profile, avatar upload  
/employees → Employee list with search/filter/CRUD [admin, hr, manager]  
/add-user → Register employee form [admin, hr]  
/projects → Project management with cards + detail view [admin, manager, developer]  
/tasks → Kanban board (todo/in-progress/review/done) with DnD [admin, manager, developer]  
/performance → Employee performance analytics [admin]  
/attendance → Punch in/out + attendance log + calendar [all]  
/leave → Apply/approve/reject leave requests [all]  
/work-logs → Daily work log entries [all]  
/support → Support tickets + real-time chat [all]  
/team-inbox → Shared team message board [all]  
/holidays → Holiday calendar with sidebar [all]  
/policies → PDF policy documents library [all]  
/emails → Email records management [admin, manager]  
/reports → Performance export reports [admin, hr]  
/settings → App + notification settings [admin, hr]  
/activity → Activity logs viewer [admin, hr, manager]

API BASE & HELPER FUNCTION (lib/api.ts)

Create apiRequest(endpoint, options) that:  
- Prefixes API\_BASE\_URL = process.env.NEXT\_PUBLIC\_API\_BASE\_URL  
- Auto-injects Content-Type: application/json (skip for FormData)  
- Auto-injects Authorization: Bearer  
- On response 401: calls showTokenExpiryModal() if not on login page

```
- Returns: { status, message, data, ok, httpStatus, total, page, totalPages }
```

```
ALL API GROUPS (export const xApi = { ... })
```

```
authApi:
```

```
login(identifier, password) → POST /user/auth/login
register(userData) → POST /user/auth/register
listUsers(params?) → GET /user/auth/users?q&role;&department;&page;&limit;
getUser(id) → GET /user/auth/users/:id
updateUser(id, payload) → PUT /user/auth/users/:id
deleteUser(id) → DELETE /user/auth/users/:id
toggleUserDisabled(id) → POST /user/auth/users/:id/toggle-disabled
updateProfile(profile) → PUT /user/auth/update-profile
uploadAvatar(file) → POST /user/auth/upload-avatar (FormData: image)
getActivityLogs(params?) → GET /user/auth/getActivityLogs
getActivityLogsById(id, params?) → GET /user/auth/getActivityLogsById?id=
```

```
passwordResetApi:
```

```
sendOtp(email) → POST /password-reset/send-otp
resendOtp(email) → POST /password-reset/resend-otp
verifyOtp(email, otp) → POST /password-reset/verify-otp
resetPassword(email, otp, newPassword, confirmPassword) → POST /password-reset/reset-password
```

```
projectsApi:
```

```
list(params?) → GET /user/projects?page&pageSize;&q;&status;&priority;
get(id) → GET /user/projects/:id
create(payload) → POST /user/projects
update(id, payload) → PUT /user/projects/:id
remove(id) → DELETE /user/projects/:id
uploadImage(file) → POST /user/projects/upload-image (FormData: image)
uploadFile(file) → POST /user/projects/upload-file (FormData: file)
uploadAudio(file) → POST /user/projects/upload-audio (FormData: audio)
uploadVideo(file) → POST /user/projects/upload-video (FormData: video)
uploadFileWithProgress(file, onProgress) → XHR POST with progress events
addAttachments(id, files) → POST /user/projects/:id/attachments
removeAttachment(id, idxOrName) → DELETE /user/projects/:id/attachments/:idx
getAttachmentUrl(filename) → returns URL string with token
```

```
tasksApi:
```

```
list(params?) → GET /user/tasks?status&priority;&projectId;&project;&page;&pageSize;&q;&user;&startDate
&endDate;&sortBy;&sortOrder;
get(id) → GET /user/tasks/:id
create(payload) → POST /user/tasks
update(id, payload) → PUT /user/tasks/:id
remove(id) → DELETE /user/tasks/:id
complete(id, done) → PUT /user/tasks/:id/complete { done }
completeForUser(id, userId, completed) → PUT /user/tasks/:id/complete-user
progress(id, progress) → PUT /user/tasks/:id/progress { progress }
approve(id, notes?) → PUT /user/tasks/:id/approve
unapprove(id) → PUT /user/tasks/:id/unapprove
toggleHold(id, hold_task, reason?) → PUT /user/tasks/:id/hold
addSubtask(id, subtask) → POST /user/tasks/:id/subtasks
updateSubtask(id, subId, patch) → PUT /user/tasks/:id/subtasks/:subId
removeSubtask(id, subId) → DELETE /user/tasks/:id/subtasks/:subId
addSubtaskAttachments(id, subId, files) → POST /user/tasks/:id/subtasks/:subId/attachments
removeSubtaskAttachment(id, subId, idx) → DELETE /user/tasks/:id/subtasks/:subId/attachments/:idx
addAttachments(id, files) → POST /user/tasks/:id/attachments
removeAttachment(id, idx) → DELETE /user/tasks/:id/attachments/:idx
addComment(id, payload) → POST /user/tasks/:id/comments
removeComment(id, commentId) → DELETE /user/tasks/:id/comments/:commentId
getComments(id) → GET /user/tasks/:id/comments
uploadImage/uploadFile/uploadAudio/uploadVideo(file) → FormData upload
getAttachmentUrl(filename) → URL string with auth token
```

```
attendanceApi:
```

```
status(userId) → GET /user/attendance/status?user_id=
punchIn(userId) → POST /user/attendance/punch-in { user_id }
punchOut(userId) → POST /user/attendance/punch-out { user_id }
todayStats() → GET /user/attendance/stats/today
todayList() → GET /user/attendance/today
checkIn() → POST /user/attendance/check-in
checkOut() → POST /user/attendance/check-out
byDate(dateIso) → GET /user/attendance/by-date?date=
```

```
range(params) → GET /user/attendance/range?from=&to;=
userHistory(userId, params?) → GET /user/attendance/user/:id?from=&to;=
myStats() → GET /user/attendance/my-stats
myWeeklyHours() → GET /user/attendance/my-weekly-hours
```

**leaveApi:**

```
my(params?) → GET /leaves/my?page&limit;&q;&month;
all(params?) → GET /leaves/all?page&limit;&q;&userId;&overviewUserId;&status;&date;&month;
apply(payload) → POST /leaves/apply { dateFrom, dateTo, reason, type?, duration? }
approve(id) → PUT /leaves/:id/approve
reject(id, reason) → PUT /leaves/:id/reject { cancellation_reason }
cancel(id, reason) → PUT /leaves/:id/cancel { cancellation_reason }
```

**supportApi:**

```
list(params?) → GET /support?page&pageSize;&status;&employeeId;&category;&assigned;_to&q;
get(id) → GET /support/:id
create(payload) → POST /support
update(id, payload) → PUT /support/:id
assign(id, assigned_to) → PUT /support/:id/assign
remove(id) → DELETE /support/:id
```

**chatApi:**

```
getMessages(ticketId) → GET /chat/:ticketId
sendMessage(ticketId, message, type) → POST /chat { ticket_id, message, username, type }
uploadImage/uploadFile/uploadAudio/uploadVideo(ticketId, file, msg?, type) → FormData
downloadAttachment(ticketId, filename, downloadName?) → fetch + blob download
getAllMessages(params?) → GET /chat?ticket_id&username;&page;&limit;
removeAttachment(messageId) → DELETE /chat/:messageId/attachment
deleteMessage(messageId) → DELETE /chat/:messageId
```

**teamInboxApi:**

```
getMessages(params?) → GET /team-inbox?page&limit;
sendMessage(message) → POST /team-inbox { message }
getMessage(id) → GET /team-inbox/:id
updateMessage(id, message) → PUT /team-inbox/:id
deleteMessage(id) → DELETE /team-inbox/:id
```

**workLogsApi:**

```
create(payload) → POST /work-logs { date, project_name?, task, blockers?, description?, time_spent?, status? }
list(params?) → GET /work-logs?user_id&from;&to;.
```

**reportsApi:**

```
getEmployeePerformance(params?) → GET
/reports/employee-performance?period&department;&role;&performance;
getPerformanceSummary(params?) → GET /reports/performance-summary?period
exportPerformance(params?) → GET /reports/export-performance?format.
```

**settingsApi:**

```
getAll() → GET /settings
updateAll(settings) → PUT /settings
getSection(section) → GET /settings/:section
updateSection(section, data) → PUT /settings/:section
```

**holidaysApi:**

```
list(params?) → GET /holidays?from&to;
create(payload) → POST /holidays (FormData or JSON: date, title, description?, image?)
update(id, payload) → PUT /holidays/:id
remove(id) → DELETE /holidays/:id
```

**emailApi:**

```
list(params?) → GET /emails?page&limit;&q;
get(id) → GET /emails/:id
update(id, payload) → PUT /emails/:id { name?, email? }
```

**Policies (raw fetch, not in api.ts):**

```
GET /policies → list policies
POST /policies → FormData: name, file, assigned_user_id?
DELETE /policies/:id → delete policy
```

**taskOtpApi:**

```
send(taskId, email?) → POST /task-otp/send-otp
verify(taskId, emails, otp) → POST /task-otp/verify-otp
recipient(taskId?) → GET /task-otp/recipient?task_id=
resend(taskId, emails?) → POST /task-otp/resend-otp
```

**DATA MODELS (TypeScript interfaces)**

```
User: { id, fullname, email, phone?, password, role, department?, address?,
  bio?, avatar_url?, website_url?, linkedin_url?, github_url?,
  date_of_birth?, employee_id?, is_disabled? }
```

```
ProjectPayload: { name, description?, status?, priority?, progress?,
  team?, startDate?, dueDate?, budget?, spent?, client?, category?,
  tasks?: { total, completed, inProgress, pending }, attachments? }
```

```
TaskPayload: { title, description?, status?, priority?, assignee?, project?,
  projectId?, startDate?, startTime?, deadlineDate?, dueDate?, progress?,
  subtasks?: Subtask[], comments?: Comment[], attachments?: string[],
  review_count?, status_history?, completed_by?, completed_at?,
  overtime_days?, user_completions? }
```

```
AttendanceRecord: { id, employee, date, checkIn, checkOut,
  status: 'present'|'absent'|'late'|'half-day', hoursWorked, userId }
```

```
LeavePayload: { dateFrom, dateTo, reason, type?, duration? }
```

```
SupportTicketPayload: { username, category?, ticket_id?, sub_category?,
  subject_sub_category?, subject, body, media?, assigned_to?,
  status?: 0|1|2|3, reseller?, message_by? }
```

```
ChatMessage: { id, username, message, ticket_id, type: 'CustomerSupport'|'User',
  created_at, attachment_type?, attachment_name?, attachment_path?, attachment_size? }
```

```
WorkLog: { id, user_id, date, project_name?, task, blockers?,
  description?, time_spent, status? }
```

```
HolidayRecord: { id, date, title, description?, image_path?,
  created_by?, created_at?, updated_at? }
```

```
Policy: { id, name, fileName, fileSize, uploadedAt, uploadedBy,
  fileUrl, assignedUser?: { id, fullname, email, employee_id? } | null }
```

```
SettingsData: { general?, notifications?, appearance?, security?, company? }
```

#### CONTEXT PROVIDERS (lib/ directory)

##### 1. TokenExpiryProvider (lib/token-expiry-context.tsx)

- showTokenExpiryModal(message?) → renders modal overlay
- Used by apiRequest() on 401 and by AuthProvider

##### 2. AuthProvider (lib/auth-context.tsx)

- State: user, token, isLoading, isAuthenticated
- login(identifier, password) → validates via API, stores to localStorage
- logout() → clears localStorage + redirects to /
- On init: reads localStorage, validates token with GET /user/auth/users
- Timeout: 3s token validation timeout before clearing

##### 3. ToastProvider (lib/toast-context.tsx)

- Provides toast notification functions app-wide

##### 4. AttendanceProvider (lib/attendance-context.tsx)

- Manages punch-in/out state for header attendance button

Wrap order in app/layout.tsx:

```
{children}
```

#### SIDEBAR NAVIGATION (components/sidebar.tsx)

Items with role restriction:

```
Dashboard /dashboard [admin, hr, manager, developer]
Profile /profile [admin, hr, manager, developer]
Employees /employees [admin, hr, manager]
Performance /performance [admin]
Projects /projects [admin, manager, developer]
Kanban /tasks [admin, manager, developer]
Attendance /attendance [admin, hr, manager, developer]
Support /support [admin, hr, manager, developer]
Team Inbox /team-inbox [admin, hr, manager, developer]
Activity /activity [admin, hr, manager]
Emails /emails [admin, manager]
Holidays /holidays [admin, hr, manager, developer]
Settings /settings [admin, hr]
Leave /leave [admin, hr, manager, developer]
Policies /policies [admin, hr, manager, developer]
```

#### REAL-TIME (Socket.IO)

Connect to: process.env.NEXT\_PUBLIC\_API\_URL (http://localhost:3005)

Used in: support-chat.tsx for live chat messages

Used in: online-members.tsx for presence detection  
Pass auth token in socket handshake for authentication

KEY FEATURES TO IMPLEMENT PER PAGE

Dashboard: stats cards (employees, present today, projects, tasks), recent activity feed, charts (attendance trends, task distribution), birthday celebrations, online members

Projects: card grid view + list view toggle, project status filter, priority filter, team member avatars, progress bar, attachment viewer, media preview (images/video/audio/files), upload with progress tracking

Tasks (Kanban): 4 columns (todo/in-progress/review/done), DnD between columns, task cards with priority badge + assignee + deadline, subtask management, comment thread, file attachments, OTP approval workflow, hold/approve actions

Support: ticket list with status filter, real-time chat panel, file/image/audio/video message attachments, emoji picker, OTP verification for ticket operations, assign to agent

Attendance: punch-in/punch-out button in header, monthly calendar heatmap, daily stats cards, late/absent/present breakdown, date range filter

FILE UPLOAD PATTERN (consistent across all modules)

1. Upload file → receive { filename, originalname, size, type } from server
2. Store server filename in attachments array on the resource
3. To display: call getAttachmentUrl(filename) which appends ?token= for auth
4. Supported types: image, file (docs), audio, video
5. For large files use XHR with onprogress event for progress bar

TAILWIND CONFIG EXTENSIONS

Extra breakpoints: xs (475px), mobile-sm (320px), mobile-md (375px), mobile-lg (425px), tablet-sm/md/lg, desktop-sm/md/lg (up to 1920px)

Extra spacing: 18 (4.5rem), 88 (22rem), 128 (32rem)

Custom animations: fade-in, slide-in, bounce-gentle, float, float-delay

Generate the full project with all files. Use realistic placeholder data for charts and dashboards. Make the UI modern, dark-mode compatible, and fully responsive down to 320px mobile width.



## 9. Quick Start & Install Commands

---

```
# 1. Create project with Next.js 15
$ npx create-next-app@latest ems-frontend --typescript --tailwind --app --no-src-dir
$ cd ems-frontend
```

```
# 2. Install all production dependencies
$ pnpm add @hello-pangea/dnd @heroui/react @hookform/resolvers
$ pnpm add @radix-ui/react-alert-dialog @radix-ui/react-avatar
$ pnpm add @radix-ui/react-dialog @radix-ui/react-dropdown-menu
$ pnpm add @radix-ui/react-label @radix-ui/react-popover @radix-ui/react-progress
$ pnpm add @radix-ui/react-select @radix-ui/react-separator @radix-ui/react-slot
$ pnpm add @radix-ui/react-switch @radix-ui/react-tabs @radix-ui/react-toast
$ pnpm add @tabler/icons-react class-variance-authority clsx
$ pnpm add cmdk date-fns embla-carousel-react emoji-picker-react
$ pnpm add framer-motion geist gsap input-otp lucide-react
$ pnpm add next-themes react-day-picker react-dropzone
$ pnpm add react-hook-form react-resizable-panels react-toastify
$ pnpm add recharts socket.io-client sonner tailwind-merge
$ pnpm add tailwindcss-animate vault zod
```

```
# 3. Create .env file
$ echo 'NEXT_PUBLIC_API_BASE_URL=http://localhost:3005/api' > .env
$ echo 'NEXT_PUBLIC_API_URL=http://localhost:3005' >> .env
```

```
# 4. Run development server
$ pnpm dev
```

This document was auto-generated from source code analysis of the EMS frontend repository. It reflects the exact API contracts, data models, routes, and configurations extracted from lib/api.ts, all app/ pages, components/, lib/ contexts, and configuration files. All model fields, enum values, and endpoint paths are production-accurate.